



# HTOC

HEALTHCARE THREAT OPERATIONS CENTER

HTOCThreatConnect — Lightweight ThreatConnect Client Package

Version 1.0  
April 2026

# Table of Contents

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>Table of Contents</b>                               | <b>3</b>  |
| <b>1. Purpose</b>                                      | <b>5</b>  |
| <b>2. Scope</b>                                        | <b>5</b>  |
| <b>3. Repository Layout</b>                            | <b>6</b>  |
| <b>4. Prerequisites</b>                                | <b>6</b>  |
| <b>5. Configuration</b>                                | <b>7</b>  |
| 5.1 config.json                                        | 7         |
| 5.2 Secrets handling                                   | 8         |
| <b>6. Installation</b>                                 | <b>8</b>  |
| 6.1 Editable install from the repository (development) | 8         |
| 6.2 Install from a wheel (distribution)                | 9         |
| 6.3 Build a wheel (maintainers)                        | 9         |
| <b>7. API Reference (High-Level)</b>                   | <b>9</b>  |
| 7.1 get_v3_threatconnect_data                          | 9         |
| 7.2 get_v2_threatconnect_data                          | 11        |
| <b>8. ThreatConnect Class (Low-Level)</b>              | <b>11</b> |
| <b>9. Smoke test (Python)</b>                          | <b>12</b> |
| <b>10. Versioning and Release Checklist</b>            | <b>12</b> |
| <b>11. Troubleshooting</b>                             | <b>13</b> |
| <b>12. Relationship to Other HTOC Components</b>       | <b>14</b> |
| <b>13. Related Documents</b>                           | <b>14</b> |

## Table of Contents

|                                           |          |
|-------------------------------------------|----------|
| <b>Table of Contents</b>                  | <b>3</b> |
| 1. Purpose                                | 4        |
| 2. Scope                                  | 4        |
| 3. Repository Layout                      | 5        |
| 4. Prerequisites                          | 5        |
| 5. Configuration                          | 6        |
| 6. Installation                           | 7        |
| 7. API Reference (High-Level)             | 8        |
| 8. ThreatConnect Class (Low-Level)        | 10       |
| 9. Smoke test (Python)                    | 11       |
| 10. Versioning and Release Checklist      | 11       |
| 11. Troubleshooting                       | 12       |
| 12. Relationship to Other HTOC Components | 13       |
| 13. Related Documents                     | 13       |

## Table of Contents 3

|                                                        |          |
|--------------------------------------------------------|----------|
| <b>1. Purpose</b>                                      | <b>5</b> |
| <b>2. Scope</b>                                        | <b>5</b> |
| <b>3. Repository Layout</b>                            | <b>6</b> |
| <b>4. Prerequisites</b>                                | <b>6</b> |
| <b>5. Configuration</b>                                | <b>7</b> |
| 5.1 config.json                                        | 7        |
| 5.2 Secrets handling                                   | 8        |
| <b>6. Installation</b>                                 | <b>8</b> |
| 6.1 Editable install from the repository (development) | 8        |
| 6.2 Install from a wheel (distribution)                | 9        |
| 6.3 Build a wheel (maintainers)                        | 9        |
| <b>7. API Reference (High-Level)</b>                   | <b>9</b> |
| 7.1 get_v3_threatconnect_data                          | 9        |
| 7.2 get_v2_threatconnect_data                          | 11       |

---

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>8. ThreatConnect Class (Low-Level)</b>        | <b>11</b> |
| <b>9. Smoke test (Python)</b>                    | <b>12</b> |
| <b>10. Versioning and Release Checklist</b>      | <b>12</b> |
| <b>11. Troubleshooting</b>                       | <b>13</b> |
| <b>12. Relationship to Other HTOC Components</b> | <b>14</b> |
| <b>13. Related Documents</b>                     | <b>14</b> |

## 1. Purpose

The **HTOCThreatConnect** folder is a small, installable Python package that provides a **read-only (GET-only)** ThreatConnect HTTP client plus two high-level helpers that return indicator data as a **pandas DataFrame**:

``get_v3_threatconnect_data`` — Queries ThreatConnect v3 [/v3/indicators](#) with TQL, paginates results, and normalizes JSON into a flat table.

``get_v2_threatconnect_data`` — Calls ThreatConnect v2 [/v2/indicators/observed](#) for a single calendar day of observed indicators per owner.

The package is intended for **scripts and internal tooling** that need a minimal dependency surface (requests + pandas + pytz) without pulling in the full official ThreatConnect SDK.

-

## 2. Scope

This procedure applies to developers and analysts who install, configure, upgrade, or troubleshoot the package from the **Location** link in the header table, or who consume it from a distributed wheel on the [Z:\](#) drive.

-

### 3. Repository Layout

| Path                                       | Role                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------|
| `HTOCThreatConnect/`                       | Main package source (`ThreatConnect.py`, `RequestObject.py`, `__init__.py`) |
| `HTOCThreatConnect/utils/config.json`      | API credentials and base URL (see Section 5)                                |
| `HTOCThreatConnect/utils/config_loader.py` | Loads and validates `config.json`                                           |
| `HTOCThreatConnect/tc.conf`                | Packaged with the wheel (see `MANIFEST.in` / `pyproject.toml` package-data) |
| `pyproject.toml`                           | Build metadata, version, dependencies                                       |
| `requirements.txt`                         | Minimal `requests` pin (full deps declared in `pyproject.toml`)             |
| `build/`                                   | Generated by `pip install` / build — safe to delete and regenerate          |

-

### 4. Prerequisites

| Requirement | Notes                                                                |
|-------------|----------------------------------------------------------------------|
| Python      | 3.9 or newer (`requires-python` in `pyproject.toml`)                 |
| Network     | Reachability to the ThreatConnect API host defined in `api_base_url` |
| Credentials | Valid `api_access_id` and `api_secret_key` in `config.json`          |

| Requirement | Notes                                                                |
|-------------|----------------------------------------------------------------------|
| Optional    | `Z:\` access if installing from a pre-built wheel under `Z:\wheels\` |

Example dependency install command:

```
pip install requests
```

-

## 5. Configuration

### 5.1 config.json

The helpers load credentials from:

HTOCThreatConnect/HTOCThreatConnect/utils/config.json

(path under the GitHub source tree linked in the header table)

When the package is installed, the same path is resolved relative to the installed package directory ([Path\( file \).parent / "utils" / "config.json"](#)).

**Required keys** (validated by `load_config`):

| Key              | Description                  |
|------------------|------------------------------|
| `api_secret_key` | ThreatConnect API secret key |
| `api_access_id`  | ThreatConnect API access ID  |

| Key               | Description                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------|
| `api_base_url`    | API root URL (e.g. `https://<host>/api`)                                                        |
| `api_default_org` | Default organization name (must match playbook/query expectations; code uses `HTOC Org` in TQL) |

If any key is missing or empty, `load_config` raises a `KeyError`.

## 5.2 Secrets handling

**Do not** commit real credentials to git. Use a local `config.json` that is excluded from version control, or inject credentials via your deployment pattern.

Rotate keys in ThreatConnect if a file is ever exposed.

Treat `config.json` like a password file: least-privilege file permissions on disk.

-

## 6. Installation

### 6.1 Editable install from the repository (development)

From your local checkout, `cd` to the folder that contains this package's `pyproject.toml` (same layout as the header **Location** link).

```
py -m pip install -e .
```

This allows editing source files and immediately using the updated package in Python.



## 6.2 Install from a wheel (distribution)

Per README.md, example:

```
py -m pip install Z:\wheels\HTOC_threatconnect-0.1.10-py3-none-any.whl
```

Upgrade an existing install:

```
py -m pip install --upgrade Z:\wheels\HTOC_threatconnect-0.1.10-py3-none-any.whl
```

Bump the version in `pyproject.toml` before building a new wheel so consumers can tell releases apart.

## 6.3 Build a wheel (maintainers)

From the same package root (`pyproject.toml` directory):

```
py -m pip install --upgrade build  
py -m build
```

Copy the generated `.whl` from [dist/](#) to [Z:\wheels\](#) (or your internal artifact location) and document the version in release notes.

-

## 7. API Reference (High-Level)

### 7.1 `get_v3_threatconnect_data`

```
from HTOCThreatConnect import get_v3_threatconnect_data
```

```
df = get_v3_threatconnect_data() # default lastObserved >= 2023-01-01
df = get_v3_threatconnect_data(lastObserved_date="2025-01-01")
```

| Parameter           | Type                                                                           | Default                | Description                                                                                |
|---------------------|--------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| `lastObserved_date` | `None`, `str`<br>(`YYYY-MM-DD`),<br>`datetime.date`, or<br>`datetime.datetime` | `None` → `"2023-01-01" | Lower bound for<br>`lastObserved` in TQL<br>(`00:00:00Z` on that<br>day)                   |
| `indicatorActive`   | `bool`                                                                         | `True`                 | When `True`, TQL<br>includes<br>`indicatorActive=true`.<br>When `False`, inactive<br>only. |
| `bothActivities`    | `bool`                                                                         | `False`                | When `True`, TQL<br>uses `(indicatorActive<br>EQ true OR<br>indicatorActive EQ<br>false)`  |

### Hardcoded behavior (in source):

**Owner:** HTOC Org only (list OWNERS in ThreatConnect.py).

**Indicator types:** Address, Email Address, File, Host, URL, ASN, CIDR, Email Subject, Hashtag, Mutex, Registry Key, Stripped URL, User Agent.

**Endpoint:** [GET /v3/indicators](#) with fields=tags,associatedGroups.

**Pagination:** resultLimit=10000, advances resultStart until a short page.

**SSL verification:** Disabled (VERIFY\_SSL = False); urllib3 insecure-request warnings suppressed.

**Output:** DataFrame with nested fields flattened via pandas.json\_normalize; deduplicated on derived indicator column; several redundant columns dropped after normalization.

## 7.2 get\_v2\_threatconnect\_data

```
from HTOCThreatConnect import get_v2_threatconnect_data

df = get_v2_threatconnect_data() # today's date (UTC) on v2 observed API
df = get_v2_threatconnect_data(dateObserved="2025-10-01")
```

| Parameter      | Type                                    | Default | Description                                                                    |
|----------------|-----------------------------------------|---------|--------------------------------------------------------------------------------|
| `dateObserved` | `None`, `str`, `date`,<br>or `datetime` | `None`  | Calendar day passed as<br>`dateObserved=YYYY-MM-DD` to v2<br>observed endpoint |

### Hardcoded behavior:

**Owner:** HTOC Org.

**Endpoint:** [GET /v2/indicators/observed?owner=...&dateObserved=...](#)

**No pagination loop** in code — assumes one response carries the full set for that day.

**SSL verification:** Disabled, same as v3 helper.

-

## 8. ThreatConnect Class (Low-Level)

Exported from `HTOCThreatConnect.ThreatConnect`:

**`read\_only = True`** — Any HTTP method other than **GET** raises `PermissionError`.

**Authentication:** HMAC (TC {accessId}:{signature}) using `api_aid` / `api_sec` as passed into the constructor.

**Retries:** Up to 5 attempts with ~59 s sleep on read timeout or connection errors.

``set_verify_ssl(bool)`` — Toggle TLS verification (helpers set it to `False` today).

Use this class only when you need custom GET paths beyond the two helpers; keep writes in separate code paths that use the official SDK or ThreatConnect UI.

-

## 9. Smoke test (Python)

After install or credential rotation, run a minimal check in the same environment as `pip install`:

```
py -c "from HTOCThreatConnect import get_v3_threatconnect_data; df =  
get_v3_threatconnect_data(); print(len(df), 'rows')"
```

A non-empty row count confirms connectivity and credentials when ThreatConnect holds matching data for the configured window; zero rows may still be valid if nothing matches TQL.

Optionally repeat for `get_v2_threatconnect_data()` with a known-good `dateObserved`.

-

## 10. Versioning and Release Checklist

Update ``version`` in `pyproject.toml` (and align wheel filename / README examples).

Run smoke tests (**Section 9**).

Build wheel (`py -m build`), publish to [Z:\wheels\](#) or internal store.

Notify consumers to `pip install --upgrade` with the new wheel path.

**Never** ship a wheel that embeds real `config.json` secrets — consumers should supply their own config or environment-specific packaging.

-

## 11. Troubleshooting

| Symptom                                                 | Likely cause                                                 | Resolution                                                                                                                  |
|---------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <code>'FileNotFoundError: Config file not found'</code> | <code>'config.json'</code> missing next to installed package | Create <code>'HTOCThreatConnect/utils/config.json'</code> with required keys; reinstall if using wheel without package-data |
| <code>'KeyError'</code> from <code>'load_config'</code> | Missing or null key in JSON                                  | Fill all four required keys                                                                                                 |
| <code>'ModuleNotFoundError: HTOCThreatConnect'</code>   | Package not installed in active interpreter                  | <code>'pip install -e .'</code> from repo root or install wheel into that environment                                       |
| <code>'PermissionError: Read-only client'</code>        | Code attempted non-GET                                       | Use official SDK or HTTP client for writes; this package is GET-only by design                                              |
| <code>'RuntimeError'</code> with binary response body   | Non-2xx API response                                         | Check URL, credentials, owner name, and TQL; decode error message from ThreatConnect                                        |
| Empty DataFrame from v3 helper                          | No indicators in window for <code>'HTOC Org'</code>          | Widen <code>'lastObserved_date'</code> or confirm data exists in ThreatConnect UI                                           |
| Empty DataFrame from v2 helper                          | No observed rows for that UTC date                           | Confirm v2 observed API returns data for the date in UI or Postman                                                          |

| Symptom            | Likely cause                                                       | Resolution                                                                       |
|--------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------|
| SSL / proxy errors | Local network or `VERIFY_SSL=False` mismatch with corporate policy | Work with network team; consider `set_verify_ssl(True)` if CA trust is available |

-

## 12. Relationship to Other HTOC Components

**Tipper, PRISM scoring, and other automation** may use either this package or the full ThreatConnect SDK under [\\10.1.4.22\\...\\threatconnect](#) — standardize on one client per project to avoid duplicate credential files.

**SOPs for playbooks** (SOP\_TC\_Playbook\_\*.md) describe ThreatConnect automation in the UI; this package is for **Python-side** read access to the same platform.

-

## 13. Related Documents

README.md alongside pyproject.toml in the `HTOCThreatConnect` source tree — quick install and usage

SOP\_PRISM\_ThreatAssessmentScoring.md — PRISM pipeline that also integrates ThreatConnect (different client path)

SOP\_Tipper\_DailyIndicatorTips.md — batch script using ThreatConnect SDK

ThreatConnect API v3 indicators: [ThreatConnect REST API](https://docs.threatconnect.com)